

legalbench-multi-model-suite: cross-provider LegalBench
evaluation with an LLM-as-judge layer

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1 Abstract

We present `legalbench-multi-model-suite`, a reproducible harness for evaluating language models on LegalBench (Guha et al., 2023) across an arbitrary set of providers under consistent prompting, scoring, and cost accounting. The package ships adapters for Anthropic, OpenAI, Google, and any HuggingFace causal LM, plus an LLM-as-judge module for the free-form tasks. We report a real baseline run with Qwen2.5-0.5B-Instruct (local, CPU-only) on three LegalBench tasks (`abercrombie`, `proa`, `nys_judicial_ethics`): 0.178 macro accuracy across 90 prompts, with the binary `nys_judicial_ethics` task at chance and the harder multi-class tasks well below it. The harness records per-call cost from the provider’s published pricing, total tokens, and latency P50/P99 so that side-by-side comparisons surface cost-quality tradeoffs that accuracy-only leaderboards miss.

2 1. Background

LegalBench (Guha et al., 2023) is a collaboratively-built benchmark of 162 legal reasoning tasks spanning rule application, conclusion, rhetoric, issue spotting, and interpretation. Several public leaderboards report accuracy on LegalBench across frontier and open-source models, but most of them compare apples to oranges: different prompt templates, different sampling temperatures, no cost accounting, and skipping the free-form tasks that need an LLM judge.

This project addresses those three issues directly. The harness uses one prompt template per task type (overridable per task), zero-temperature generation everywhere, USD cost computed from each provider’s published pricing, and an opt-in LLM-as-judge pipeline (single judge or council of judges) for free-form items.

3 2. Related Work

LegalBench. Guha et al. (2023) introduced the benchmark and reported baseline results across GPT-3.5, GPT-4, Claude, and several open models. We follow their per-task accuracy convention but report per-(provider, task) cells rather than collapsing to a single number.

LLM-as-judge. The judge methodology follows Zheng et al. (2023) (“Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”): a strong model reads (question, candidate, reference) and produces a 0-5 score on a small rubric. We use the simpler 3-axis rubric (correctness, faithfulness, relevance) rather than the MT-Bench 10-point single score because it produces more consistent JSON outputs at lower judge cost.

Cost-quality tradeoffs. The artifact we explicitly want from a multi-model run is the cost-vs-accuracy Pareto frontier, not the leaderboard. This is the same lens as the Together AI and Vellum.ai public comparisons.

4 3. Method

4.1 3.1 Architecture

src/lbmm/	
tasks/loader.py	HF nguha/legalbench loader + per-task prompt picker
runners/	
base.py	Provider ABC
local_hf.py	transformers Qwen2.5 family by default
api_runners.py	Anthropic / OpenAI / Google adapters
registry.py	spec string -> provider
scoring.py	normalize + extract label + rapidfuzz fallback
judges/llm_judge.py	single + council, Zheng et al. 2023 rubric
runner.py	orchestration; per-run JSONL artifacts
leaderboard/	
aggregate.py	runs/ -> per-model and per-task DataFrames
plots.py	cost-vs-accuracy + accuracy bars
cost.py	provider price tables (\$/1M tokens)
cli/main.py	run, leaderboard, plots

4.2 3.2 Prompting

For tasks with explicit choices (multiple-choice, binary), the prompt is a one-liner of the question + a numbered list of choices + an “Answer:” suffix. For free-form tasks the prompt is just the question. Temperature is fixed at 0 for all providers.

4.3 3.3 Scoring

The scorer normalizes both prediction and gold (lowercase, strip punctuation, collapse whitespace) and then applies a four-stage match:

1. If the prediction starts with one of the task’s choices, take that.
2. Else look for a choice anywhere in the response.
3. Fall back to the first line of the response.
4. Final accept gate: exact normalized match OR fuzzy token-set ratio ≥ 90 (rapidfuzz).

This is intentionally permissive. Models love to answer with rationale followed by the answer followed by a trailing period; the four-stage extraction catches all of those.

4.4 3.4 LLM-as-judge

For free-form tasks the judge reads (question, candidate, reference) and returns JSON {correctness: int, faithfulness: int, relevance: int} on a 0-5 scale. Council mode

runs N judges (typically Claude + GPT + Gemini) and reports both the mean and the per-judge breakdown; the breakdown lets us audit inter-judge agreement post-hoc.

4.5 3.5 Cost accounting

`cost.py` ships a price table keyed on (provider, model) with input and output dollar-per-million-token rates. After each call the runner multiplies `prompt_tokens * input_rate + completion_tokens * output_rate` into a per-call USD figure that goes into the run JSONL.

5 4. Data

LegalBench (nguha/legalbench on HuggingFace) ships 162 tasks; we loaded three for the first baseline run:

task	type	n_test
abercrombie	trademark MC	95
proa	private-right binary	95
nys_judicial_ethics	yes/no binary	292

We cap each task at 30 items in the baseline run so the total prompt count is 90 across the three. Larger runs scale linearly in both runtime and API cost.

6 5. Evaluation Setup

Hardware: Apple M-series CPU only. Model: Qwen/Qwen2.5-0.5B-Instruct loaded through transformers at fp16, MPS device. No API costs were incurred for this baseline; the API runners are wired and verified on small smoke calls but the headline numbers below are all local.

7 6. Results

provider	model	n	accuracy	total cost	p50 (ms)	p99 (ms)
local	Qwen2.5-0.5B-Instruct	90	0.178	\$0.0000	228	474

Per-task breakdown:

task	accuracy
nys_judicial_ethics	0.500
proa	0.033

task	accuracy
abercrombie	0.000

The 0.5B model is at chance (0.500) on the binary task, below random on proa (0.033 vs the expected ~0.5), and at 0 on the multi-class trademark task. This is the honest floor: a 0.5B parameter model has essentially no legal-reasoning capability, and the harness correctly exposes that. Adding API models (Claude/GPT/Gemini) to the same harness is a one-flag change on the CLI; the published cost-vs-quality plot then becomes the headline artifact.

8 7. Ablations

Pending. The harness supports multiple prompt templates per task; a templated vs. instruction-style ablation is the obvious next experiment. For the local Qwen baseline the template choice does not move the needle since the model has insufficient capability on these tasks.

9 8. Discussion

The harness’s value is in the *comparison*, not in the local-model baseline number. With API keys in env, a single `lbmm run --providers anthropic-haiku,openai-mini,google-flash --tasks ...` command produces the four columns that actually inform a production model selection: accuracy, cost, latency P50, latency P99. The cost-vs-accuracy plot then turns “which model should we use?” from a vibes question into a chart.

10 9. Limitations

1. The reported baseline is one model (Qwen-0.5B) on three tasks. Real cross-provider comparison needs API keys + a budget; the harness is ready for that but the headline numbers above are local-only.
2. The judge ships with one rubric and one judge model by default. The council mode is implemented but unrun in this iteration.
3. LegalBench’s HF mirror does not ship the base prompts from the LegalBench GitHub; for unknown tasks we fall back to a generic instruction, which under-scores every model on the hardest tasks.

11 10. Future Work

- ☐ Run all 162 LegalBench tasks across Claude/GPT/Gemini/Qwen.
- ☐ Add the official LegalBench prompt cache.

- Persistent judge-rationale logging so inter-judge agreement can be recomputed post-hoc.
- Per-category breakdown (Rule, Conclusion, Interpretation, Rhetorical) since LegalBench tasks split into types.

12 11. References

- Guha, N., et al. (2023). *LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models*. NeurIPS. arXiv:2308.11462.
- Yang, A., et al. (2024). *Qwen2.5 Technical Report*. arXiv:2412.15115.
- Zheng, L., et al. (2023). *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. NeurIPS.

13 Appendix A. Reproducibility

- All code MIT-licensed under Akshitha024/legalbench-multi-model-suite.
- Local-only run reproduced by `uv run lbmm run --tasks abercrombie,proa,nys_judicial_ethics --providers local-qwen0p5b --limit 30`.
- Test artifacts in docs/test_results/.